

Calculators, cell phones, or notes are not allowed. You must show all your work and explain how you obtain your answer. Just guessing the correct number will not give any credit.

- (1) a) Use Euclian algorithm to compute the GCD and LCM of 54 and 42.
 b) Find all integer solutions to $54X + 42Y = 8$.
 c) Find all integer solutions to $54X + 42Y = 12$.

Solution: a) $54 = 42 + 12$, $42 = 12 \cdot 3 + 6$, $12 = 6 \cdot 2 + 0$ so the GCD is 6 and the LCM is $54 \cdot 42 / 6 = 378$. b) Since 6 does not divide 8, this equation has no solutions. c) The equation is equivalent to $9X + 7Y = 2$. The solutions are $(-6 + 7k, 8 - 9k)$ for any $k \in \mathbb{Z}$.

- (2) Compute $\varphi(105)$ and 11^{4804} modulo 105.

Solution: $105 = 3 \cdot 5 \cdot 7$ and so $\varphi(105) = 2 \cdot 4 \cdot 6 = 48$. We have $11^{48} = 1$ and thus $11^{4804} = 11^4 = 121^2 \cong 16^2 \cong 256 \cong 46$.

- (3) Solve $X \cong_{25} 4$ and $X \cong_{103} 8$ for some $0 \leq X < 2575$. (You do not need to compute the final answer eg $22 + 17 \cdot 26$ is ok)

Solution: From the first eqn we have $X = 4 + 25k$ and from the second eqn $4 + 25k \cong_{103} 8$ so that $25k \cong_{103} 4$. We now compute the inverse of 25 in $\mathbb{F}_{103}$. $103 = 4 \cdot 25 + 3$ and $25 = 3 \cdot 8 + 1$ so $1 = 25 - 8(103 - 4 \cdot 25) = 33 \cdot 25 - 8 \cdot 103$ i.e. $33 \cdot 25 \cong_{103} 1$. But then $k \cong 33 \cdot 25k \cong 33 \cdot 4 \cong 132 \cong 29$. Finally $X = 4 + 25 \cdot 29$.

- (4) Show that $\mathbb{F}_{17}[i]$ is **not** a field and find $(2 + 3i)^{-1}$.

Solution: We have $4^2 + 1^2 = 17$ so $(4 + i)(4 - i) = 17 \cong 0$ so $4 + i$ is a 0 divisors (but fields never have zero divisors). On the other hand $N(2 + 3i) = 2^2 + 3^2 = 13$ is invertible modulo 17 (since $13 \cdot 4 = 1 + 3 \cdot 17$) and so $(2 + 3i)^{-1} = (2 - 3i)/13 = (2 - 3i)4 = 8 - 12i = 8 + 5i$.

- (5) The powers of 2 mod 11 are given below.

2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}
2	4	8	5	10	9	7	3	6	1

Use discrete logarithms with base 2 to find **all** solutions of $6x^4 = 8$ in $\mathbb{Z}/11\mathbb{Z}$.

Solution: Taking the discrete logarithm $I(\dots)$ we obtain $I(6) + 4I(x) = I(6x^4) = I(8)$ and so $9 + 4I(X) = 3$ so that $4I(X) = -6 \cong_{10} 4$, so $I(X) = 1, 6$ so $X = 2, 9$.

- (6) a) List the order of each element of the group $(\mathbb{Z}/7\mathbb{Z})^\times$.

a	1	2	3	4	5	6
order a						

- b) Which of the elements are primitive 6-th roots of 1 ?

- c) Use b) to compute $\Phi_6(X)$.

Solution: The order of 1 is 1, the order of 2 is 3 (since $2^i = 2, 4, 8 \cong 1$), the order of 3 is 6 (since $3^i = 3, 9 \cong 2, 3^3 \cong 6, 3^4 \cong -3, 4^5 \cong -2, 4^6 \cong -6 \cong 1$), the order of 4 is 3 (since $4^i = 4, 16 \cong 2, 8 \cong 1$), the order of 5 is 6 (since $5^i = 5, 25 \cong 4, 20 \cong 6, -5, -4, -6 \cong 1$). The primitive roots are 3 and 5. Thus $\Phi_6(X) = (X-3)(X-5) = X^2 - 8X + 15 = X^2 - X + 1$.

- (7) Compute the cyclotomic polynomial $\Phi_{10}(X)$.

Solution:

$$\Phi_{10}(X) = \frac{X^{10} - 1}{\Phi_1(X)\Phi_2(X)\Phi_5(X)} = \frac{X^{10} - 1}{(X^5 - 1)\Phi_2(X)} = \frac{X^5 + 1}{X + 1} = X^4 - X^3 + X^2 - X + 1.$$

- (8) Does $X^2 + 4X - 15 = 0$ have a solution in $\mathbb{F}_{103}$?

Solution: The above equation is equivalent to $(X+2)^2 = 19$. We compute

$$\left(\frac{19}{103}\right) = -\left(\frac{103}{19}\right) = -\left(\frac{8}{19}\right) = -\left(\frac{2}{19}\right)^3 = 1$$

(where the first equality holds as $19, 103 \cong_4 3$ and the last as $2 \cong_8 3$ so that $(\frac{2}{19}) = -1$). Therefore the above equation has a solution.

- (9) Factor $14 + 5i$ in to indecomposable elements of $\mathbb{Z}[i]$ (note: $14^2 + 5^2 = 221 = 13 \cdot 17$).

Solution: $(14 + 5i) \cdot \overline{(14 + 5i)} = N(14 + 5i) = 14^2 + 5^2 = 221 = 13 \cdot 17$. Since $13 = 2^2 + 3^2$ and $17 = 1^2 + 4^2$, the possible factors are $3 \pm 2i$ and $4 \pm i$. We now compute

$$\frac{14 + 5i}{4 - i} = \frac{(14 + 5i)(4 + i)}{17} = \frac{51 + 34i}{17} = 3 + 2i.$$

The factorization is then $14 + 5i = (4 - i)(3 + 2i)$. (Or $(-4 + i)(-3 - 2i)$ or $(1 + 4i)(-2 + 3i)$ or $(-1 - 4i)(2 - 3i)$.)

- (10) The continued fraction algorithm for $\alpha > 1$ is purely periodic of period 2 such that $[\alpha] = [\alpha_2] = [\alpha_4] = 10$ and $[\alpha_1] = [\alpha_3] = [\alpha_5] = 1$, thus

$$\alpha = 10 + \frac{1}{1 + \frac{1}{10 + \dots}}.$$

Find α . Using this, find all solutions to $X^2 - 35Y^2 = 1$.

Solution: We have $\alpha = 10 + \frac{1}{1 + \frac{1}{\alpha}} = 10 + \frac{\alpha}{\alpha + 1} = \frac{11\alpha + 10}{\alpha + 1}$ and so $\alpha^2 - 10\alpha = 10$ i.e. $(\alpha - 5)^2 = 35$ i.e. $\alpha = 5 + \sqrt{35}$ (as $\alpha > 10$). We now compute

$$5 + \sqrt{35} = 10 + \frac{1}{1 + \frac{1}{5 + \sqrt{35}}} = 10 + \frac{5 + \sqrt{35}}{6 + \sqrt{35}}, \quad \text{so}$$

$$\sqrt{35} = 5 + \frac{5 + \sqrt{35}}{6 + \sqrt{35}} = \frac{6\sqrt{35} + 35}{\sqrt{35} + 6}$$

which tells us that the first solution is $(6, 1)$. All other solutions are given by $X_k + \sqrt{35}Y_k = (6 + \sqrt{35})^k$.